

Conditional LoRA Bridges at 230M: What Replicates, What Softens, and What Inverts

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2026-07-03

Abstract

Our controlled study of conditional LoRA bridges [1] established its findings on a purpose-trained 33.6M-parameter model, leaving open which conclusions are properties of the mechanism and which are artifacts of scale. We port the unchanged experiment suite to a pretrained 230M-parameter model (LiquidAI LFM 2.5) and rerun it at both the standard and the 10× training budget. The methodological core replicates exactly: bits-per-byte cannot detect sensor conditioning (the capacity-matched constant-feature control matches the true-feature bridge, +0.0852 vs +0.0856), while task-aligned probes detect it decisively (0.72 rank-1 on six-way IMU, 0.31 on fifty-way audio, all controls at chance). Three scale-dependent results reverse or soften. Prefix tuning flips sign, from -0.0048 BPB on the small model to +0.0614 at 230M, while remaining the weakest adapter. The diversity-BPB tradeoff softens: the strongest penalty retains 89% of the unregularized BPB gain where the small model retained none. Most strikingly, the long-budget collapse inverts: extended unregularized task training destroyed IMU conditioning on the small model (rank-1 0.80 to 0.17, chance) but strengthens it at 230M (0.72 to 0.89) — even as weight collapse under pure text loss persists at both scales. At the two scales tested, capacity and pretraining are allies of task-conditioned weight-space bridges, and single-scale claims about the mechanism’s failure modes should be read as lower bounds.

1 Motivation

The conditional-LoRA-bridges study [1] reported one sharp methodological result — aggregate language-modeling metrics cannot certify sensor conditioning — and one mechanism result, hyper-network weight collapse, with facets that include degradation from token-space (prefix) conditioning and a budget-dependent destruction of task-level conditioning under prolonged unregularized training. All of it was measured on a single 33.6M-parameter model trained from scratch. This note answers the obvious question: which of those findings survive contact with a real pretrained model?

2 Porting the harness

We add a checkpoint adapter (`hf:<model-id>`) to the published harness so that every experiment — baselines, controls, probes, budgets, seeds — runs unchanged against a HuggingFace causal LM; here LiquidAI LFM 2.5 230M [2], a 14-layer hybrid short-convolution/attention model with a 65,536-token vocabulary. Three porting details matter for fidelity:

Table 1: BPB improvement over the frozen baseline (mean across audio, vision, IMU; standard budget; seed-42 protocol as in the original paper). The decomposition — bridge $\approx$ constant control < static LoRA — replicates; prefix tuning flips sign.

Method	33.6M (paper)	230M (this note)
Static LoRA	+0.0033	+0.0876
Conditional bridge	+0.0027	+0.0856
Shuffled features	+0.0018	+0.0853
Random features	+0.0006	+0.0823
Constant features	+0.0027	+0.0852
Prefix tuning	-0.0048	+0.0614

- **LoRA targets.** The paper’s target set “all” (every attention and MLP linear) maps to q/k/v/out projections on the six attention layers and the SwiGLU projections on all fourteen layers; short-convolution projections are left unadapted. The generated LoRA vector has $D = 774,144$ dimensions at rank 4.
- **Byte-exact BPB across tokenizers.** Bits-per-byte is tokenizer-independent only if the per-token byte table reconstructs the exact UTF-8 length of any encoded text. We build and validate such a table for the model’s byte-level BPE, including its non-special added tokens, so BPB values are comparable across the two vocabularies.
- **Sequence-start sensitivity.** LFM 2.5 is strongly BOS-dependent (7.13 nats/token without BOS vs. 2.70 with, on the same text). The packing dataloader already places BOS at every row start; the task probes prepend BOS for `hf`: models so the probe measures conditioning, not an out-of-distribution artifact. The published small-model protocol is unchanged and reproduces bit-identically.

The frozen 230M model evaluates at 1.1839 BPB on the paper’s validation stream (the small model: 1.9654).

3 What replicates

Table 1 gives the BPB decomposition at the standard budget. The paper’s central methodological claim transfers unchanged: the conditional bridge (+0.0856) is indistinguishable from the capacity-matched constant-feature control (+0.0852), static LoRA is the best pure-text adapter (+0.0876), and three-seed spreads are below 10^{-3} . BPB still cannot see conditioning; the task probes still can (Table 2): true features reach 0.72 rank-1 on six-way IMU and 0.31 on fifty-way audio while shuffled, random, and no-bridge controls sit at chance — numerically close to the small model’s 0.80 and 0.33. Composition of independently trained bridges remains sub-additive at both scales, and unregularized text-loss training still collapses generated weights toward input-independence (cross-input cosine 0.99 at $\lambda = 0$).

4 What softens

Prefix tuning flips sign. On the small model, feature-conditioned prefix tuning strictly degraded BPB (-0.0048); at 230M it improves it (+0.0614) — yet remains the weakest adapter by

Table 2: Task-probe rank-1 accuracy for true features vs. the no-bridge reference, at both budgets. The small model’s IMU probe collapses to chance at the 10× budget; the 230M model’s strengthens.

Probe	Budget	Chance	33.6M		230M	
			true	none	true	none
IMU	standard	0.17	0.80	0.17	0.72	0.16
IMU	10×	0.17	0.17	0.17	0.89	0.16
AUDIO	standard	0.02	0.33	0.03	0.31	0.02
AUDIO	10×	0.02	0.36	0.03	0.61	0.02

Table 3: Cross-input cosine of generated weights (IMU; lower = more input-dependent) across the diversity sweep, at both scales and budgets.

λ	33.6M std	33.6M 10×	230M std	230M 10×
0.00	0.97	0.99	0.99	0.98
0.01	0.92	0.99	0.98	0.98
0.05	0.65	0.31	0.96	0.98
0.10	0.44	0.47	0.60	0.68
0.20	0.55	0.17	0.72	0.36

a wide margin against the $\sim +0.0856$ weight-space rows. The original claim is therefore correctly scoped to small models trained without prefixes, while the paper’s ordering — weight-space conditioning dominates token-space conditioning — holds at both scales.

The diversity–BPB tradeoff softens. On the small model, enforcing input-dependent weights at $\lambda = 0.20$ erased the BPB gain entirely; at 230M the same penalty retains 89% of it ($+0.0767$ of $+0.0860$). The non-monotonicity the paper reports at the largest tested weight (cross-input cosine reversing at $\lambda = 0.20$) appears at the standard budget at 230M as well (0.60 at $\lambda = 0.10$ vs. 0.72 at $\lambda = 0.20$), but at the 10× budget the reversal disappears and cross-input cosine falls to 0.36 at the largest weight (Table 3). The penalty required for input-dependence grows with both budget and scale: on the small model $\lambda = 0.01$ loses its effect at the long budget, while at 230M penalties below $\lambda = 0.10$ never produce input-dependent weights at either budget.

5 What inverts

The sharpest warning in the original paper was budget-dependent: at 10× training, the small model’s unregularized IMU bridge became behaviorally indistinguishable from no bridge at all (rank-1 0.17, equal to chance), having scored 0.80 at the standard budget. At 230M the same protocol *strengthens* conditioning: IMU rank-1 rises from 0.72 to 0.89 (top-2 1.00), and audio rises from 0.31 to 0.61 against a 0.02 chance level, with every control still at chance. Prolonged task-aligned training is an ally of conditioning at this scale, not a failure mode. The weight-space collapse under pure text loss still occurs at 230M (Table 3, $\lambda = 0$); what changes is that label-paired training at scale maintains input dependence without regularization over the budgets tested.

6 Implications

Three practical updates for anyone building on the paper. First, the evaluation discipline (capacity-matched controls, task-aligned probes) transfers verbatim and remains necessary: BPB is exactly as blind at 230M as at 33.6M. Second, the mechanism’s failure modes shift in the favorable direction from the 33.6M model to the 230M one — the task-conditioning collapse threshold moves outward and the cost of enforcing input-dependent weights shrinks. Two caveats bound that statement: the comparison confounds scale with pretraining (the small model was trained from scratch, LFM 2.5 is pretrained), and it concerns task-conditioned training only — collapse under pure text loss is unchanged at 230M. Even so, claims about weight-space conditioning’s fragility measured at one scale should be read as lower bounds. Third, token-space and weight-space conditioning both improve with scale, but the gap between them persists; weight-space modulation remains the stronger interface across every configuration tested.

Reproducibility

The port is additive: the published harness reproduces the original paper bit-identically, and the same runner produces every number here via `--checkpoint hf:LiquidAI/LFM2.5-230M`. Artifacts: the standard-budget suite (35 experiments) and the 10× suite (33 of 35; the two remaining seed-stability repetitions were not run) ship alongside this note as `summary-lfm.json` and `summary-lfm-scaling.json` in the code repository, with the original paper’s artifacts unchanged. Code: <https://github.com/kortexa-ai/legolm.paper>.

References

- [1] F. Penov. Conditional LoRA Bridges for Modular Sensor Adaptation of Frozen Small Language Models. Preprint, 2026. <https://github.com/kortexa-ai/legolm.paper>.
- [2] Liquid AI. LFM2.5-230M. Model release, 2025–2026. <https://huggingface.co/LiquidAI/LFM2.5-230M>.